



Parkinson's disease tremor prediction towards real-time suppression: A self-attention deep temporal convolutional network approach

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ABSTRACT

Accurate prediction of Parkinson's disease tremor (PDT) is crucial for developing assistive technologies; however, this is challenging due to the nonlinear, stochastic, and nonstationary characteristics of PDT, which substantially vary among patients and their activities. Moreover, most models only have one-step prediction capabilities, which causes delays in real-time applications. This paper proposes a self-attention deep temporal convolutional network (SADTCN) model for the real-time prediction of hand-arm PDT signals from different activities and joint angular motions. The SADTCN can capture both short- and long-term dependencies and complex temporal and spatial dynamics of PDT signals and hence, can effectively adapt to varying tremor characteristics. The performance of the proposed model is evaluated using experimental hand-arm PDT data. The results show that the SADTCN outperforms existing deep learning (DL) models by accurately predicting varying tremor amplitudes and frequencies multi-step ahead. Moreover, we performed spectrum analysis on the measured and predicted signal using the short-time Fourier transform (STFT) as a measure of potential active tremor control and found that SADTCN can accurately determine the transience of tremor amplitude in frequency and time. Finally, we run the Wilcoxon signed-rank statistical test and the results show a statistically significant improvement in the proposed model over the other DL models in all conditions. Therefore, the SADTCN can overcome the nonstationary, nonlinear, and stochastic nature of PDT to perform multi-step prediction with high accuracy, robustness, and generalizability in unseen testing data.

1. Introduction

Parkinson's disease (PD) is a neurodegenerative movement disorder characterized by the progressive loss of dopaminergic and other subcortical neurons in deep parts of the brain, with uncertain etiology [1]. After Alzheimer's disease, PD is the second most prevalent age-related neurodegenerative ailment, affecting approximately 2% of people over 60 years and more than 3% of people over 75 years of age [2]. PD is usually identified by its motor symptoms and thus, it is diagnosed clinically [3]. The crucial features of a PD diagnosis are bradykinesia, tremors, rigidity, postural instability, and non-motor symptoms like insomnia, anxiety, and fatigue [4]. Specifically, PDT (about 70% of PD cases) is a prominent upper-limb motor symptom of numerous neurological movement diseases; characterized by involuntary and pseudo-rhythmic movements in the frequency range of 3 to 8 Hz (with multiple harmonics) that disrupt coordination, precision, and velocity of desired motions [5].

Currently, the treatment of PD patients depends on dopaminergic and anticholinergic medications and deep brain stimulation (DBS) [6, 7]. However, it is reported that medications are often ineffective for the treatment of PDT and have adverse side effects [8], while no evidence-based therapy guidelines exist concerning the efficacy of specific dopaminergic medicines [9]. On the other hand, DBS has significant risks and can have a very high cost per patient [10]. As a potential alternative to treating PDT, assistive technologies such as wearable devices have recently sparked significant interest due to their promising performance in compensating for tremulous movements with negligible side effects [11]. However, the tremor signals exhibit highly nonlinear, stochastic, and nonstationary characteristics, which substantially vary among patients and their activities. Therefore, for designing assistive devices effectively and suppressing PDT in real-time, accurate prediction of PDT signals is crucial [12,13].

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In the literature, various adaptive filtering and machine learning techniques were proposed to estimate tremors. Early research developed the Fourier linear combiner (FLC) for tremor estimation using a least mean square (LMS) algorithm, assuming that the tremor signal is periodic with specified frequency [14]. Nevertheless, the tremor signal is not fixed and periodic but is time-varying, nonlinear, and stochastic. The weighted-frequency Fourier linear combiner (WFLC) was designed to address the shortcomings of the FLC, which models quasi-periodic signals as a modulating sinusoid and utilizes a modified LMS algorithm to track its frequency, amplitude, and phase [15]. Nonetheless, the performance of WFLC is impaired if the tremor signal has numerous dominant frequencies and the frequency changes very fast. The band-limited multiple Fourier linear combiner (BMFLC) and double adaptive BMFLC were developed to solve the problem of estimating tremor signals with many dominant frequencies [16,17]; however, they operate with a predefined set of frequencies, and when more frequencies are combined to improve estimation accuracy, the computational load increases. Thus, the Extended-BMFLC (EBMFLC) was developed, which outperformed its predecessors [18]. Yet, the drawback of the EBMFLC and the earlier works is that they introduce an intrinsic temporal delay for tremor control in wearable assistive devices [19].

Adaptive least squares support vector machine (SVM) was proposed to estimate hand tremor with nonlinear kernels [20]. Although it outperformed FLC algorithms, the high computational cost limits its use in real applications. Some other studies developed tremor prediction models using autoregressive (AR) and autoregressive-moving average (ARMA) [21]. However, these models performed poorly. The WAKE framework based on wavelet decomposition and adaptive Kalman filtering (KF) was introduced [22] and some other studies developed KF techniques to estimate tremor [23,24]. Nevertheless, they are limited to one-step prediction and high processing time. Due to the difficulties and shortcomings of earlier works, deep learning-based approaches were recently introduced, which demonstrated robustness in learning from data and high estimation accuracy. Deep learning methods have been used for accurately predicting various medical signals, e.g., electrocardiogram (ECG) [25] and electroencephalogram (EEG) [26], which further motivated this research in predicting PDT signals.

Recently, a PHTNet framework was developed by incorporating deep bidirectional recurrent neural networks to predict pathological hand tremor [27]. However, the model incurred a high computational load due to a complex learning architecture and extensive training. Yoon et al. [28] proposed a signal decomposition and ensemble method with deep neural networks to predict voluntary motions only without estimating tremulous motions. Zanini et al. [29] predicted Parkinson's disease electromyography (EMG) signals using multilayer perceptron (MLPs) and recurrent neural network models. Pascual-Valdunciel et al. [30] proposed long short-term memory neural networks (LSTM) to predict tremor signals. Nevertheless, the works [27–30] are limited to one step ahead prediction only. Ibrahim et al. [31] used a one-dimensional convolutional-multilayer perceptron model for multi-step prediction of tremor in rest fixed hand posture and tasks related to holding and moving pencils. The findings showed that a high average estimation accuracy was attained using 18 subjects, but only for voluntary motions. However, tremulous motions differ greatly from voluntary motions in terms of their characteristics. Moreover, the above-mentioned methods are not generalized and cannot estimate a large number of PDT conditions for standard activities examined clinically.

In this paper, we develop a self-attention deep temporal convolutional network (SADTCN) model for multi-step prediction of hand-arm PDT signals. In the first stage, the temporal convolutional network (TCN) discovers the temporal interactions within the PDT signal. Then, a self-attention layer processes and enhances the context correlatively by appropriately assigning weights based on different sequence segments. Another TCN layer then refines and tunes them before passing them to the final dense MLP layers to make the final prediction. The novelty of the SADTCN lies in its unique architecture and it is the

first to be applied and optimized for tremor signal prediction. The proposed model can capture both short- and long-term dependencies and complex temporal and spatial dynamics of PDT signals and hence, can adapt to varying tremor characteristics effectively. Although self-attention mechanisms have been applied in other fields, e.g., activity detection [32] and load forecasting [33], these applications do not possess specific challenges inherent in PDT signal prediction, such as handling the nonlinear and nonstationary tremor characteristics with short and long term dependencies and the necessity for real-time processing.

To obtain a robust model, the SADTCN model is trained and validated using experimental hand-arm PDT data from 320 samples of 20 PD patients. Then, the performance of the proposed SADTCN is evaluated using 192 test samples from 12 PD patients and compared to the current DL models, namely the standard deep neural network (DNN), long short-term memory (LSTM), and convolutional neural network (CNN). The tested activities are drinking, outstretching, resting, and winging. It is found that the proposed model outperforms the existing DL models by accurately predicting a broad range of tremor amplitudes and frequencies for a variety of activities and joint angular motions multi-step ahead with different horizons, which is crucial for developing assistive technologies. Moreover, such accurate predictions can assist clinicians in tremor assessment and decision-making, leading to improved patient care.

The highlights of this paper are summarized as follows:

1. Unlike existing DL models that are limited to one-step prediction of PDT or applicable for limited activity, this paper proposes a SADTCN model for predicting PDT multi-step for various activities and joint angular motions with varying tremor characteristics. This is the first work investigating the proposed problem and achieving enhanced generalizability through more comprehensive testing conditions.
2. Compared to current DL models, the proposed SADTCN model can better capture the nonlinear and nonstationary characteristics of PDT signals, leading to more accurate predictions of tremulous motions in real-time. In the proposed SADTCN, the TCN enables a larger effective receptive field, which fully utilizes the prior data, and the self-attention layer assigns each data point to a weight based on its importance in predicting the subsequent data that significantly influences the prediction performance.
3. The spectrum analysis on the measured signal and SADTCN-predicted signal using STFT shows that the proposed model can accurately determine the transience of tremor amplitude in both frequency and time. Such accuracy is crucial to effectively control the PDT.
4. From the Wilcoxon signed-rank test, we observed significant differences in the prediction performance of our proposed SADTCN model and other DL models in all conditions. Therefore, the proposed model can fully capture the diverse PDT characteristics from different activities, further demonstrating this study's contribution.

2. Methods

2.1. Experimental protocol and procedures

This study was conducted under the experimental protocol number NMRR-14-1694-21740 (IIR) in the Neurology Clinic of Penang General Hospital with the approval of the Medical Research Ethics Committee, Secretariat of National Institutes of Health, Malaysia. Guidelines from the National Institution of Health were followed to conduct the study. In this clinical study, 63 PD patients participated in the measurement and each patient engaged in four distinct activities: drinking, outstretching, resting, and wing. Upon receiving approval from the International Parkinson and Movement Disorder Society, the upper limb

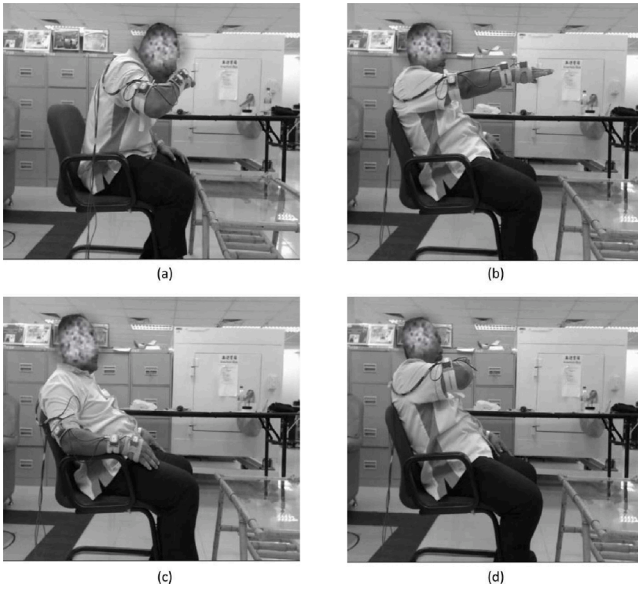


Fig. 1. Experimental setup to record hand-arm PDT during (a) drinking, (b) outstretching, (c) resting, and (d) wing activities. The patient's face is blurred due to a privacy concern.

Table 1

The main features of the PD dataset used in this study.

Dataset features	Values
Patient characteristics:	
Total number of subjects	32
Age, years (mean $\pm$ std)	69.1 $\pm$ 7.7
Gender (% female)	35%
Hoehn and Yahr stage ^a	
I/II/III/IV (% total subjects)	38.9/27.8/27.8/5.6
Recordings:	
Total samples	512
(Drinking/Outstretching/Resting/Wing)	(128/128/128/128)
Device output sampling rate (Hz)	100 Hz

^a The Hoehn and Yahr stage is a commonly used system for assessing the severity of Parkinson's disease [37].

resting and outstretching postures were carried out under the guidelines in the Movement Disorder Society's revised Unified Parkinson's Disease Rating Scale (MDS-UPDRS). The drinking and wing activities were carried out according to the guidelines of the Washington Height-Inwood Genetic Study of Essential Tremor (WHIGET) rating scale (wTRS) [34].

2.2. Data description and preparation

An experimental setup was developed to record hand-arm PDT data as shown in Fig. 1. Each patient carried out four distinct activities, such as (i) the upper limb was rested on the arm for 15 s, (ii) the upper limb was outstretched in front of the chest for 15 s, (iii) the arm was held in wing position for 15 s, and (iv) a cup was taken and water was drunk before the cup was returned to the original position. The main features of the dataset are given in Table 1. To measure the three-dimensional (3D) orientation of each hand-arm joint, three modules of attitude and heading reference system (AHRS) (model IG-500A, SBG Systems, Rueil-Malmaison, France) made up of micro-electromechanical systems (MEMS) were employed on the back of the hand, the upper arm, and the lower arm to record tremor data. The recorded quaternion data were used to calculate the relative hand-arm motions defined in terms of joint angles [35,36], such as wrist abduction-adduction (WAA), wrist flexion-extension (WFE), elbow flexion-extension (EFE), and elbow pronation-supination (EPS).

Then, we applied a bandpass filter (zero-phase 4th order Butterworth) between 3–30 Hz to extract the tremor signal, which is termed as the joint angular displacement $\Delta\theta_{\text{joint}}$. Following that, the data is split into training, validation, and test sets by choosing a random set of patients with each patient having different activity and joint angular motions. Specifically, the recorded data were divided into training, validation, and test sets with 288 samples (18 patients), 32 samples (2 patients), and 192 samples (12 patients), respectively. Note that each patient comprises 16 samples (4 activities $\times$ 4 joint angular motions) and each sample represents a patient, one activity, and one joint angular motion.

2.3. Proposed SADTCN model

In recent decades, convolutional neural networks have been widely used in various applications. In most tasks, convolutional networks outperform recurrent networks (which are designed for sequence-to-sequence learning), paving the way for the next generation of convolutional networks, the temporal convolutional network, which has a larger effective memory. In this paper, we propose a SADTCN model that can capture the nonlinear, stochastic, and spatiotemporal characteristics of PDT signals. Specifically, the proposed SADTCN can learn the underlying structure in a sequence generated by the dilated convolutional module, allowing it to capture a broader range of inputs and long-range dependencies of the historical data. The overview of the proposed SADTCN model is illustrated in Fig. 2, which consists of three main components: TCNs, self-attention layers, and dense layers.

With PDT signals, the input sequence typically represents a segment of the signal recorded from a PD patient during an activity, and the output sequence represents the predicted values that the model aims to learn. In particular, the model learns a mapping from the input sequence to the output sequence to make accurate predictions of PDT signals. We apply self-attention to the TCN, allowing the model to further extract necessary information by measuring the significance of data at various time steps. The self-attention layer can capture the spatial features of the tremor signal, while TCNs can capture the temporal characteristics with their dilated convolutions. However, the representations learned in different layers of a TCN may be abstract and high-level. By feeding high-level representations into a dense network, the model can further learn complex temporal dependencies. Thus, the dense network is used to capture the temporal characteristics further. The TCN is composed of 1-D convolutional modules with dilation factors of $D = [2^n]$, $n = 0 \dots 5$. The detailed 1-D convolutional module is shown in Fig. 3. Each convolutional module is a residual block comprising two layers of dilated convolutional and the rectified linear unit (ReLU) activation function. The nodes in them can hierarchically capture and extract information from high-order neighbors by stacking them together. Finally, the fully connected output layer would map the output of the previous layers to the prediction of the tremor signal.

The proposed SADTCN model has the following desirable features:

- **Hierarchical feature learning:** The combination of TCNs and self-attention allows for hierarchical feature learning. While TCNs excel at capturing temporal patterns, self-attention can efficiently capture spatial relationships between different elements in the input sequence. This hierarchical approach can help the model capture both short-term and long-term dependencies in the tremor signal effectively.
- **Multi-step prediction:** The TCN blocks capture long-range temporal dependencies essential for forecasting data points in future time steps. Self-attention layers dynamically shift the focus to the most relevant historical information, enhancing prediction accuracy. The dense layer synthesizes these insights into precise multi-step forecasts. Overall, each component combined enables nuanced understanding and prediction of sequential data over multiple future points, making the model highly effective for complex multi-step prediction tasks.

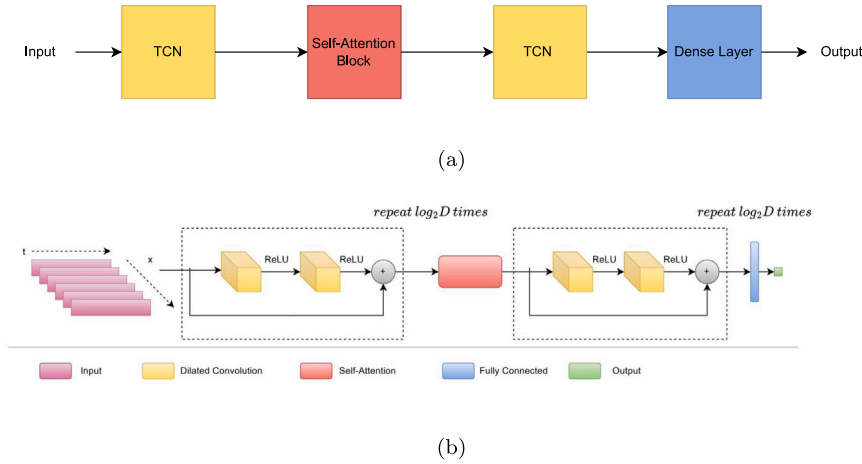


Fig. 2. The proposed SADTCN model. (a) Main components and (b) an overview of the network architecture.

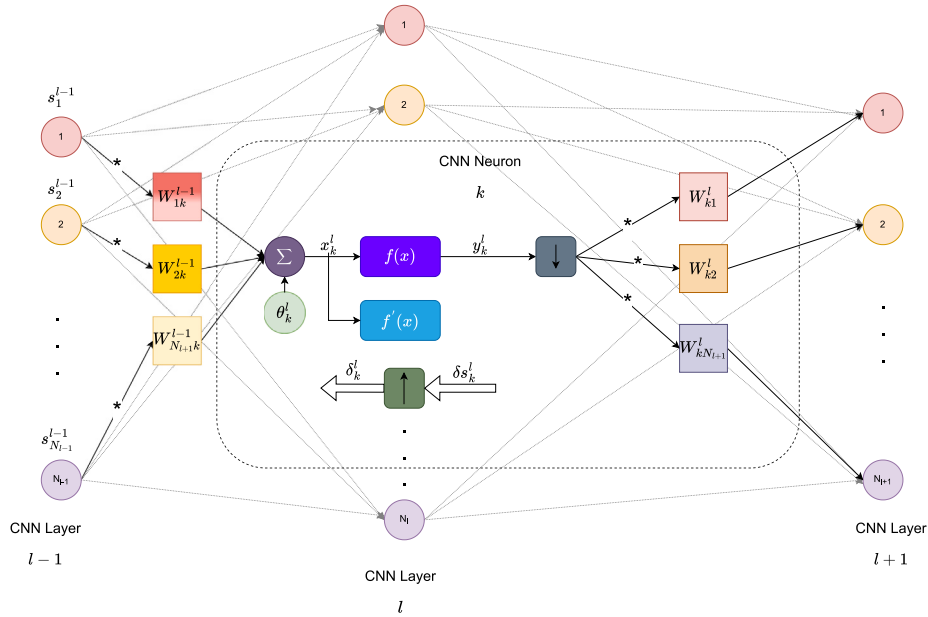


Fig. 3. The detailed operation of the 1-D convolutional module.

- **Enhanced contextual information:** Self-attention can provide enhanced contextual information by assigning appropriate attention weights to different parts of the sequence, which allows the model to focus on more relevant parts of the sequence when making predictions. This can be especially beneficial in cases where specific segments of the tremor signal are more informative.
- **Reduced vanishing gradient problem:** TCNs are known for their ability to mitigate the vanishing gradient problem, which can occur in deeper neural networks. Using TCNs in combination with self-attention can help stabilize the learning process and facilitate the training of deeper architectures.
- **Scalability:** The combination of TCNs and self-attention can be scalable to handle longer tremor signal sequences, making it suitable for practical applications that involve continuous analysis over extended periods.

2.3.1. Dilated causal convolution

The TCN uses dilation in the convolutional filter to enhance the receptive field. To enhance the receptive field size, large layers are required, which is challenging when using the model in computationally intensive tasks. To circumvent this issue, dilated convolutions allow

for an exponentially wider receptive field whilst using fewer layers. Therefore, using large dilation enables the output of higher layers to encompass a wider range of inputs. With an input sequence of $\{x_0, \dots, x_T\}$ and an output prediction sequence of $\{\hat{y}_0, \dots, \hat{y}_T\}$, the causal convolutional layer restricts the input observation to be $x_0, \dots, x_t$, which can be considered as inputs when predicting y_t at time t . In real-time applications, causal convolution ensures that there will be no leakage of historical information when predicting future data points. The receptive field R_{field} can be enhanced by using a larger dilation factor $d \in D$ and a larger filter size K_{size} or the number of stacks of residual blocks N_{stack} as

$$R_{field} = 1 + 2(K_{size} - 1) \cdot N_{stack} \cdot \sum_i d_i, \quad (1)$$

Distinctly, for the 1D input sequence $x \in \mathbb{R}^T$, where T is the temporal dimensionality, define a discrete function $g : \mathbb{Z}^+ \mapsto \mathbb{R}$ such that $g(s) = x_s$. A temporal convolution $*_d$ where d denotes a dilation factor, is expressed as follows [38]

$$H_s = (g *_d w)(s) = \sum_{i=0}^{k-1} w(i) \cdot x_{s-di}, \quad s \in [1, T] \cap \mathbb{Z}^+, \quad (2)$$

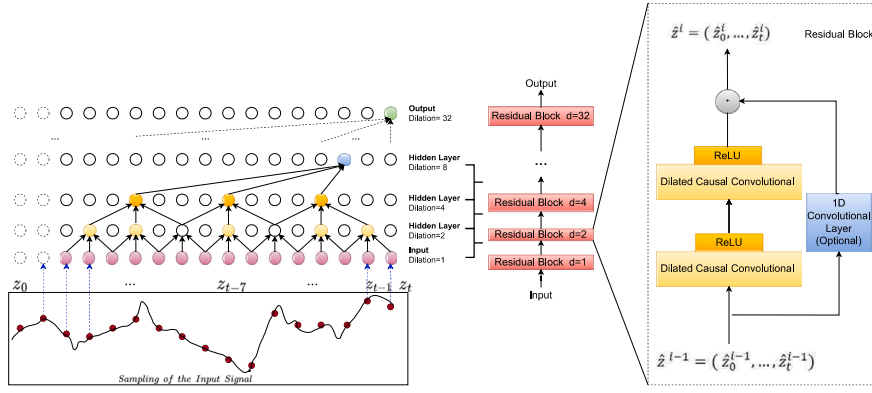


Fig. 4. Architecture of a residual block made up of two blocks. Each block contains a dilated causal convolutional layer and a ReLU activation function.

where H_s represents the s th element of the 1D output signal $H \in \mathbb{R}^T$ and $w : \{0, \dots, k-1\} \mapsto \mathbb{R}$ is the size of the 1D discrete filter, and $s-d_i$ accounts for the direction to the past.

2.3.2. Residual network

The vanishing gradient problem and optimization difficulties are the two most challenging tasks in very deep network training. The TCN layer composed of a few residual blocks doubles the receptive field width and allows very deep networks to be stabilized by facilitating gradient flow. Moreover, skip connections are used in the residual network because they can lead to faster network convergence, which speeds up the training process [39].

The architecture of a residual block is illustrated in Fig. 4, which is made up of two blocks, each containing a dilated causal convolutional layer/network (with the same dilation factor) and a ReLU activation function. The output of the two dilated causal convolutional blocks is combined with the input of the current residual block and used as the input for the subsequent residual blocks. The input and output of the residual block are $\hat{z}^{i-1} = (\hat{z}_0^{i-1}, \dots, \hat{z}_t^{i-1})$ and $\hat{z}^i = (\hat{z}_0^i, \dots, \hat{z}_t^i)$, respectively.

2.3.3. Self-attention layers

The attention block aims to develop a one-to-one association between all temporal moments. The self-attention mechanism can simultaneously operate on all elements of the input sequence and measure the relative importance of data from other timestamps with the current input. This parallel processing allows the model to attend to different parts of the tremor signal simultaneously, capturing spatial relationships efficiently. Although there are relevant works combining the self-attention layer with LSTM and generative adversarial network (GAN), self-attention has been less frequently investigated in the context of TCN. The proposed SADTCN model learns to detect global, long-range dependencies effectively within the data sequence by incorporating self-attention into TCN. Given an input sequence $x_i = \{x_0, \dots, x_T\} \in \mathbb{R}^{d_x}$, the corresponding output sequence $\hat{y}_i = \{\hat{y}_0, \dots, \hat{y}_T\} \in \mathbb{R}^{d_y}$ is generated by applying a weighted sum of transformations to input elements x_j based on the calculated attention score. The i th output $\hat{y}_i$ is

$$\hat{y}_i = \sum_{j=0}^T \alpha_{ij} (W_v x_j), \quad (3)$$

where α_{ij} denotes the weight coefficient and $W_v \in \mathbb{R}^{d_x \times d_y}$ is the trainable parameter matrices. The relevance between i th and j th input elements is calculated by executing a scaled dot product of the respective input parameter matrices using the following function [40]:

$$\alpha_{ij} = \text{Attention}(Q, K, V) = \text{softmax} \left(\frac{Q \cdot K^T}{\sqrt{d_k}} \right) V \quad (4)$$

where Q , K , and V respectively denote the Query, Key, and Value, and d_k is the dimension of the matrix K . To prevent the dot products from being too large when the dimension of d_k is high, they are scaled down by dividing it by $\sqrt{d_k}$. After obtaining the weight coefficient (from the matching level of Q and K), the output of the matching function is passed through the Softmax function, which represents the probability mapping function that generates the attention distribution. The result is thus multiplied with V to extract relevancy from each data point towards the current target data point to be predicted and obtain the final attention value.

2.4. Performance measures

For a quantitative analysis of the performance, the root mean square error (RMSE) between the actual PDT signal and the predicted signal is calculated as [28,31]

$$\text{RMSE} = \sqrt{\frac{\sum_{t=1}^N (y(t) - \hat{y}(t))^2}{N}}, \quad (5)$$

where N is the length of the signal, $y(t)$ is the actual tremor signal, and $\hat{y}(t)$ is the predicted tremor signal at time t .

To determine whether there are any significant differences between the predictions of the proposed model and those of other DL models, we statistically compare prediction scores (PS) (that come from the same patients) using the Wilcoxon signed-rank test [41].

$$PS = 2 \left\{ 1 + \sum_{i=r}^{i=q} \sum_{j=1}^i \prod_{j=n}^{n=r-q} \left[(r-q-n+1) \prod_{q=1}^{q-2+n} \right] \right\} / \frac{|2q|}{|q \times |q|} \quad (6)$$

where $\prod_j^i$ denotes the number of j -part partitions of i , r is the serial number of possible rank totals, q is the number of replicates, and n is an integer denoting the serial number of the term in the data series.

2.5. SADTCN network implementation

The proposed model is trained and validated to generate a robust and generalizable model using experimental hand-arm PDT data. To train the model, the MSE between the predicted signal and ground truth is used as a loss function. A validation loss is also employed to monitor the model's performance during the training process. The parameters are learned using a mini-batch gradient descent algorithm and Adam optimizer, with a learning rate of 0.0005. To increase the receptive field, a dilation factor of $D = [2^n]$, $n = 0 \dots 5$ is used. The TCN employs 128 filters with a filter size of 3. The model is trained using 300 epochs with a training batch size of 32 and is validated using a validation batch size of 64 to balance computational efficiency and stability during the loss evaluation. The train-validation-test dataset is processed using the sliding window technique to prepare the input of the model. Causal padding (which pads the input with additional zeros before the input

sequence) is used to predict data in early time steps of the frame, which is particularly beneficial for processing with sequential data. The early stopping technique is used to reduce the likelihood of overfitting and the training duration when the training process is not improved or the validation loss does not improve continuously for 10 epochs. The dropout layer was excluded from the residual block as it decimates early state information due to an exponential decay over the sequence length, which has a long sequence. To get better results, it is not necessary to include a normalization layer (e.g., batch normalization and weight normalization) within the residual block. The pooling layer is excluded because it may reduce the temporal resolution of features collected by convolutional layers.

2.6. Baseline models implementation

We consider three baseline models for comparison, e.g., CNN [31], DNN [28], and LSTM [30] models. The CNN model implementation consists of two stages: feature extraction and prediction. The feature extraction stage consists of a 1D convolutional layer with 128 filters and a kernel size of 3. Causal padding is used to predict the values of early time steps. Pooling layers are not used as adding maximum and average pooling layers may impair the performance of the model after rounds of experiments. A flattened layer is used to reduce the dimensionality of the feature map into a 1D vector and feed into the prediction stage. For the prediction stage, 3 fully connected layers with ReLU activation function between each dense layer are used. The CNN model is trained using the MSE loss function and Adam optimizer. The DNN model consists of 7 fully connected layers and a ReLU function is used between each layer. The DNN model is trained using the Adam optimizer and the MSE loss function. The LSTM model is designed with the notion that it will learn from a large sequence of data with multiple time steps. The training dataset is initially converted into a 3D array with samples, time steps, and feature format. The model contains 2048 blocks and employs a hidden layer of 5 neurons to generate the predicted output. The ReLU function is used as the activation function rather than the *tanh* function for higher performance. The LSTM model is trained using the Adam optimizer with MSE as the loss function. All models are trained for 200 epochs using early stopping and validation loss criteria. The CNN, DNN, and LSTM models use training batch sizes of 64, 32, and 32 and validation batch sizes of 64, 64, and 256, respectively, to achieve their best performance.

3. Results

3.1. Performance evaluation

The experimental data are split into training, validation, and testing sets. Then, we evaluate the performance of all models using the testing data. Figs. 5, 6, 7, and 8 illustrate the performance of DL models on the testing dataset for 5-step ahead prediction of PDT for different activities and joint angular motions, concerning the measured tremulous motion (i.e., the ground truth as discussed in Section 2.2). The inset graphs show how accurately the DL models can predict PDT signals with abrupt changes in amplitude/frequency. It is also crucial to accurately determine the tremor amplitude in both frequency and time to effectively control the PDT. Hence, we run spectrum analysis on the measured tremor signal and the SADTCN predicted signal using short-time Fourier transform (STFT) over the operation time as a measure of active tremor suppression. The spectrum contents of the signals (i.e., tremor occurrence time and frequency) are captured smoothly using a Hamming window of size 300. Specifically, Figs. 5(a), 6(a), 7(a), and 8(a) show the performance of DL models for 5-step ahead prediction of hand-arm PDT motions of WAA during drinking activity, EPS during outstretching activity, WFE during resting activity, and EPS during wing activity, respectively, and Figs. 5(b), 6(b), 7(b), and 8(b)

respectively show the corresponding spectrum of the measured and SADTCN predicted signals using STFT.

The results show that the SADTCN model can more accurately predict the PDT motions multi-step ahead compared to the current DL models (e.g., DNN, LSTM, and CNN) for all tested activities and joint angular motions. Besides, the spectrum analysis shows that the proposed model can predict the transience of tremor amplitude in both frequency and time. The performance of deep learning models for 5-step ahead prediction is given in Table 2. It is found that for all activities and joint angular motions, the proposed model has the lowest prediction error (RMSE) compared to the benchmark DL models.

To further demonstrate the performance of the proposed SADTCN model, we run the Wilcoxon signed-rank statistical test to evaluate how significantly the predictions of the proposed SADTCN model vary compared to other DL models. The absolute difference between the ground truth (measured signal) and the prediction results of each model are used for pairwise comparison. Table 3 gives the statistical test results for different activities and joint angular motions. The p-values are found to be less than 0.05 for all cases, further supporting the fact that there are significant differences between the proposed SADTCN model and other DL models in terms of prediction performance.

3.2. Characteristics of PDT tested

The characteristics of PDT vary significantly from patient to patient and different activities. Our PDT testing dataset consists of PDT signals from different activities and joint angular motions that exhibit diverse amplitude and frequency characteristics as given in Table 4. Such diversity is crucial to assess the prediction ability of DL models and develop a generalized model. For example, the joint angular motion (WAA) during drinking activity (Fig. 5) shows high vibration amplitude and low frequency with the dominant frequency occurring at about 8.3 s. In Fig. 6, the joint angular motion (EPS) during outstretching activity exhibits very low vibration amplitude and low/medium frequency with the dominant frequency appearing at about 3.6 s. For resting activity (Fig. 7), the joint angular motion (WFE) shows low or very low vibration amplitude and high frequency with the dominant frequency appearing at about 8.2 s, whereas for wing activity (Fig. 8), the joint angular motion (EPS) exhibits medium vibration amplitude and high frequency with the dominant frequency occurring at about 7.2 s.

Fig. 9 illustrates the overall performance (5-step ahead prediction) of deep learning models over 192 test samples for different activities and joint angular motions. For convenience, the RMSE of four different joint angular motions, i.e., WAA, WFE, EPS, and EFE are averaged for individual patients and their activities. It is found that the proposed SADTCN model exhibits the smallest prediction error among the study DL models for all activities. Furthermore, the proposed model gives more consistent results than other baseline DL models (which gave high RMSE in some cases). Therefore, the proposed model can be employed as a generalized model for predicting PD tremors for different activities and joint angular motions.

3.3. Predictability

Multi-step prediction is essential to reduce the delay in assistive devices [19]. Therefore, we analyze the predictability of DL models (over a time horizon) using the sliding window technique. Table 5 gives the mean prediction error for predicting joint angular motions over 224 samples (192 samples from 12 test subjects and 32 samples from 2 validation subjects) for different future time steps. It is found that the error increases gradually from a one-step to a ten-step ahead prediction. For one-step ahead prediction, the SADTCN outperformed the DNN and LSTM; however, the CNN marginally outperformed the SADTCN. On the other hand, the SADTCN substantially outperformed the other DL models for 3-step, 5-step, and 10-step ahead predictions.

Note that the processing time (computational cost) required by the proposed system for 5-step ahead prediction is about 20 ms, while the prediction time for real-time applications of physical assistive devices is typically considered within 20–100 ms [31,42]. The ability of the

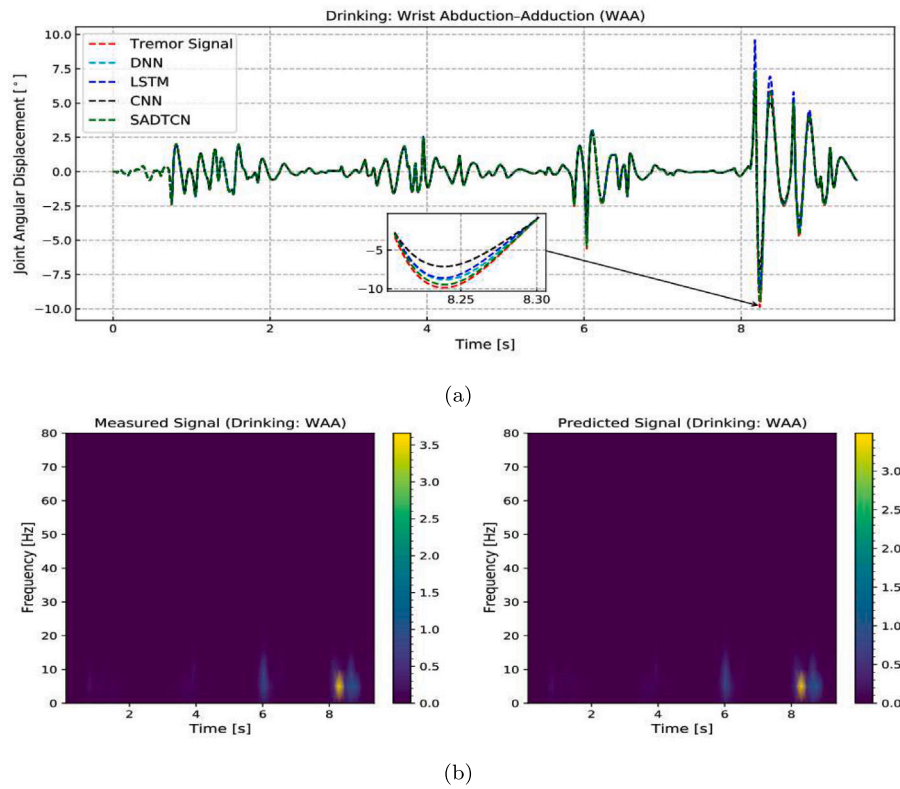


Fig. 5. (a) Performance of deep learning models in predicting PDT signals during drinking activity and (b) spectrum of measured and SADTCN predicted signals.

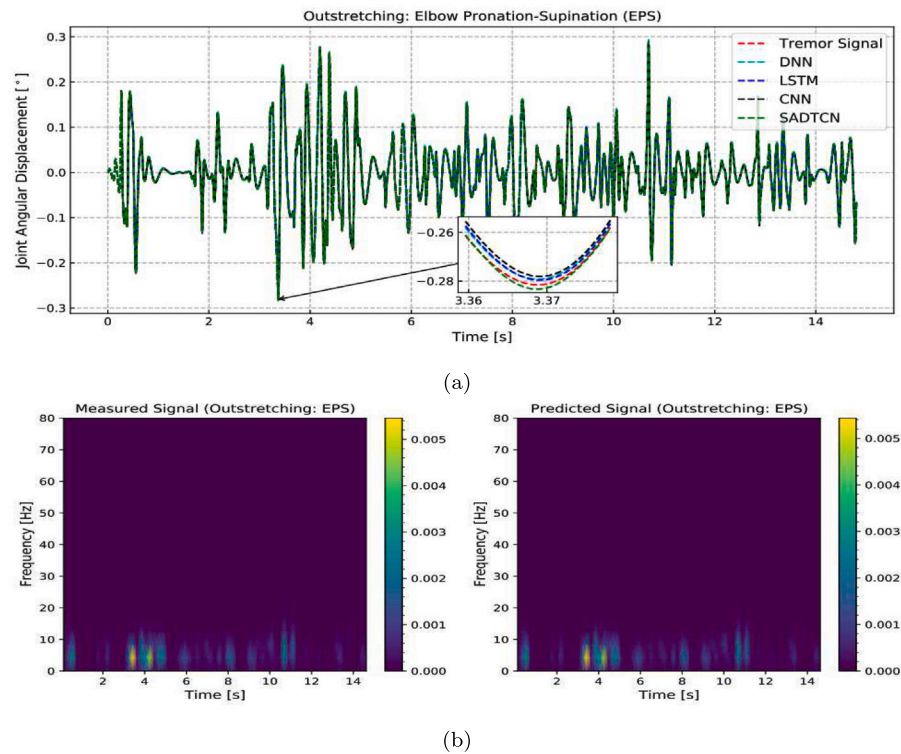


Fig. 6. (a) Performance of deep learning models in predicting PDT signals during outstretching activity and (b) spectrum of measured and SADTCN predicted signals.

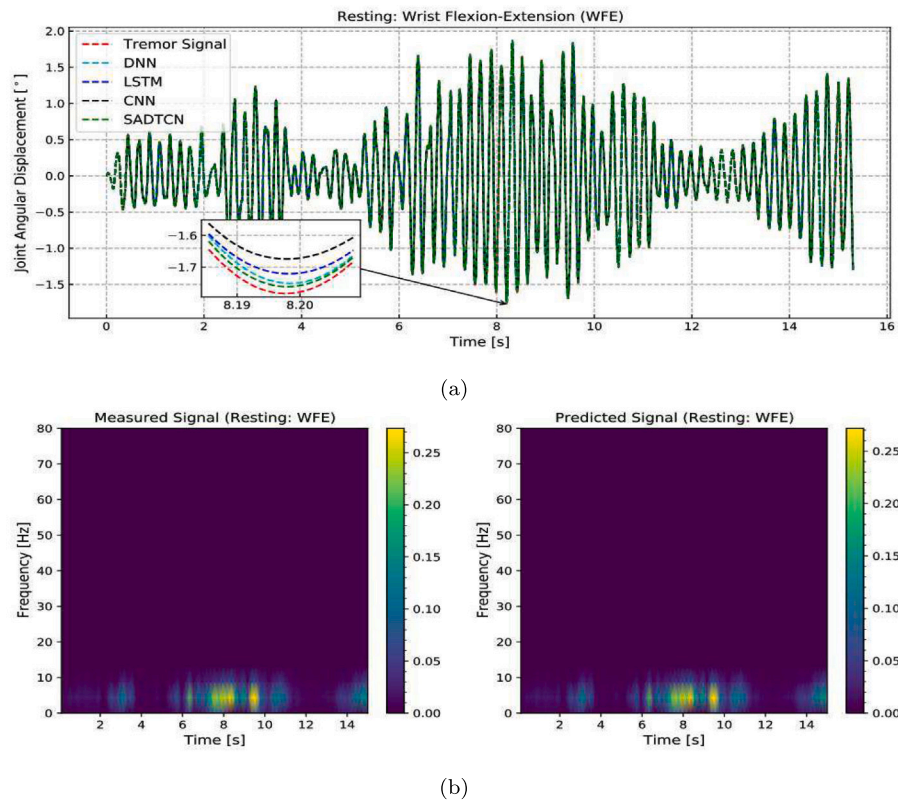


Fig. 7. (a) Performance of deep learning models in predicting hand-arm PDT signals during resting activity and (b) spectrum of measured and SADTCN predicted signals.

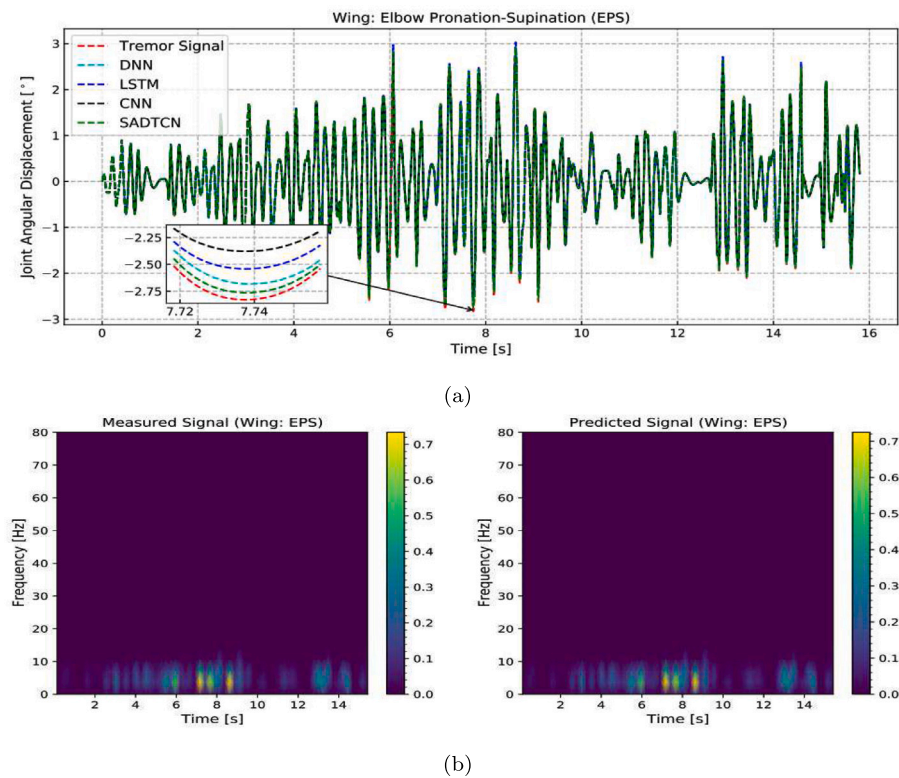


Fig. 8. (a) Performance of deep learning models in predicting hand-arm PDT signals during wing activity and (b) spectrum of measured and SADTCN predicted signals.

Table 2

Performance of 5-step ahead prediction of deep learning models for different activities and joint angular motions.

Activity	Model	Joint angular motions [°]			
		WAA	WFE	EPS	EFE
Drinking, Outstretching	SADTCN	0.041, 0.001	0.005, 0.003	0.061, 0.001	0.047, 0.001
	DNN	0.099, 0.002	0.009, 0.005	0.151, 0.002	0.111, 0.002
	LSTM	0.206, 0.002	0.017, 0.006	0.621, 0.003	0.293, 0.002
	CNN	0.253, 0.002	0.024, 0.006	0.407, 0.002	0.293, 0.002
Resting, Wing	SADTCN	0.001, 0.018	0.006, 0.009	0.004, 0.015	0.002, 0.002
	DNN	0.002, 0.032	0.014, 0.016	0.008, 0.025	0.004, 0.005
	LSTM	0.002, 0.049	0.008, 0.014	0.007, 0.042	0.004, 0.004
	CNN	0.003, 0.044	0.010, 0.021	0.012, 0.056	0.005, 0.006

Table 3

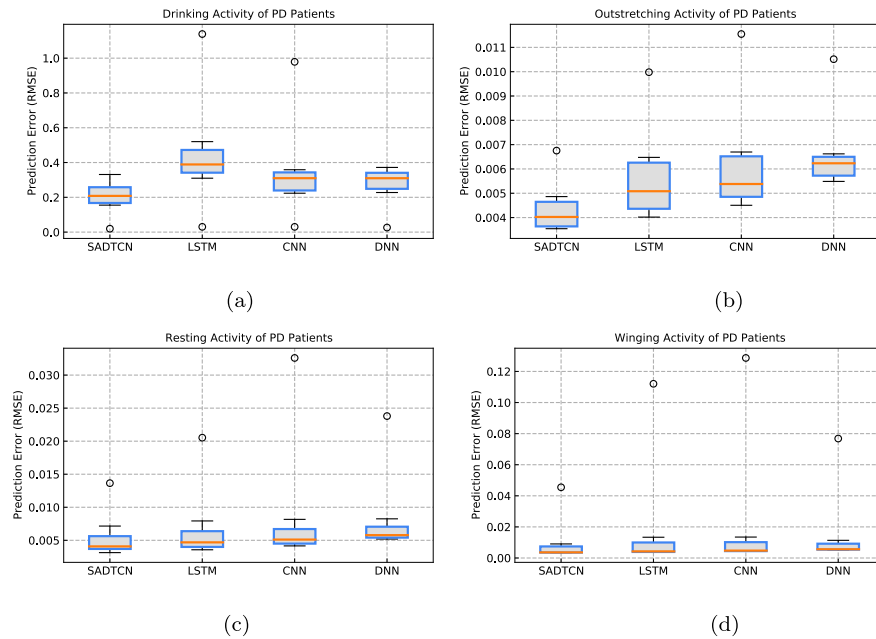
Wilcoxon signed-rank statistical test prediction scores for different activities and joint angular motions.

Activity	Actions	P-values		
		DNN	CNN	LSTM
Drinking, Outstretching	WAA	<0.01, <0.001	<0.05, <0.01	<0.001, <0.001
	WFE	<0.001, <0.001	<0.01, <0.01	<0.05, <0.01
	EPS	<0.05, <0.01	<0.05, <0.05	<0.001, <0.01
	EFE	<0.001, <0.01	<0.05, <0.01	<0.001, <0.05
Resting, Wing	WAA	<0.001, <0.05	<0.05, <0.05	<0.01, <0.05
	WFE	<0.05, <0.05	<0.01, <0.05	<0.01, <0.05
	EPS	<0.05, <0.001	<0.01, <0.01	<0.05, <0.01
	EFE	<0.001, <0.05	<0.01, <0.05	<0.01, <0.05

Table 4

Characteristics of PDT tested (n = 192 samples): range of tremor amplitude and average peak frequency.

Activity	Amplitude range [°], Peak frequency [Hz]			
	WAA	WFE	EPS	EFE
Drinking	−10.70–14.41, 3.88	−1.89–2.15, 3.40	−5.23–5.34, 4.08	−4.81–5.36, 3.45
Outstretching	−4.45–3.45, 7.32	−5.19–4.82, 7.04	−7.88–8.49, 6.54	−1.60–2.09, 5.01
Resting	−1.62–1.51, 4.49	−3.31–2.96, 4.65	−3.38–2.66, 4.68	−1.79–1.74, 4.48
Wing	−2.68–2.53, 5.39	−4.00–3.96, 5.18	−3.23–4.10, 4.69	−2.14–2.06, 4.72

**Fig. 9.** Overall performance of deep learning models over 192 test samples. The orange line indicates the median, the box indicates the range between 25%–75% quartiles, and the black lines indicate the standard deviation range.

proposed model for 5-step predictions at once can greatly reduce the processing time of the algorithm. Thus, our proposed system has the potential to meet real-time requirements by considering the sensing time, processing time of the algorithm (computational complexity), and hardware reaction time, as it is expected to be within the 100 ms range.

4. Discussion

For clinical practice, accurate PDT prediction is crucial to identify/understand the severity of tremors that arise from different daily activities, while to develop assistive devices effectively for real-time

Table 5
Performance of deep learning models for multi-step ahead prediction of test and validation subjects in different horizons.

Activity	Time horizon	Mean prediction error (RMSE)			
		SADTCN	DNN	LSTM	CNN
Drinking, Outstretching	1	0.044, 0.005	0.086, 0.028	0.131, 0.031	0.042, 0.005
	3	0.079, 0.007	0.128, 0.035	0.188, 0.044	0.099, 0.016
	5	0.133, 0.015	0.161, 0.042	0.211, 0.049	0.168, 0.034
	10	0.328, 0.031	0.350, 0.054	0.392, 0.057	0.353, 0.049
Resting, Wing	1	0.004, 0.006	0.020, 0.022	0.014, 0.024	0.002, 0.005
	3	0.006, 0.008	0.026, 0.026	0.022, 0.025	0.012, 0.020
	5	0.018, 0.012	0.033, 0.029	0.035, 0.031	0.027, 0.024
	10	0.035, 0.028	0.052, 0.036	0.049, 0.039	0.048, 0.033

tremor suppression, accurate PDT prediction is important. It is evident from Figs. 5–8 and Table 1 that the proposed SADTCN model performed the best amongst the DL models, while the spectrograms demonstrate that the model can predict the transience of tremor amplitude in both frequency and time for various activities and joint angular motions. The statistical results show that there are significant differences between the predictions of the proposed model and those of other DL models. From Fig. 9(a), it is found that the proposed SADTCN produced the lowest prediction error for drinking activity and all joint angular motions. The performance of CNN and DNN is comparable, whereas the LSTM performed the worst. For outstretching activity as shown in Fig. 9(b), the SADTCN performed the best among the DL models, while the DNN produced the worst result, and the LSTM performed better than the CNN. For resting activity (Fig. 9(c)), the SADTCN outperformed other DL models. The LSTM performed marginally better than the CNN and DNN, although the CNN slightly outperformed the DNN. For wing activity as shown in Fig. 9(d), the SADTCN outperformed other DL models, while the LSTM, CNN, and DNN yielded similar results.

The predictability from Table 2 shows the applicability of the proposed SADTCN model for practical deployment. The RMSE for DNN, LSTM, and CNN for 10-step ahead prediction for drinking activity increased by 6.7%, 19.5%, and 7.9%; for resting activity increased by 48.6%, 22.9%, and 37.1%; for outstretching activity increased by 58.8%, 67.6%, and 44.1%; and for wing activity increased by 28.6%, 39.3%, and 17.9%, respectively, as compared to the SADTCN model. Hence, the proposed SADTCN model shows the best performance for multi-step predictions of all activities and joint angular motions. This is because the proposed model can better learn the underlying and long-term relationships, and spatiotemporal characteristics from the PDT data.

Although it is proven that LSTM is capable of capturing sequential information because of its recurrent connection, with the same capacity, the TCN in SADTCN can achieve a longer effective memory than LSTM [38] and avoid the vanishing gradient issue common in RNN-based models like LSTM. The CNN can only handle fixed-size inputs, losing the ability to recognize dependencies between various positions in a sequence, which TCN can handle due to the stacking of numerous dilated convolutional layers. Additionally, CNN is less capable than TCN to fully utilize historical data as it can only review history with a linear size proportionate to the depth of the network, as seen in the increased error rate. In contrast, the DNN loses the spatial feature of the data as it is unable to capture sequential information in the input data, which prevents it from comprehending the complex relationships in the time-series sequence, resulting in higher prediction errors than SADTCN across all activities.

The TCN in SADTCN is not susceptible to the selection of hyperparameters if the receptive field size is enough, which greatly reduces the uncertainty in the search for a better model, whereas LSTM is sensitive to the various random weight initializations. One of the key reasons that the SADTCN outperformed different state-of-the-art deep learning models is that the self-attention layer in the SADTCN learns and provides more information by giving varying weights of their relevance based on the impact of the data used in predicting the current

output. Lastly, sandwiching the self-attention layer between the TCN blocks serves as a latent vector transformation mechanism for the first TCN block and offers more valuable insights to the subsequent TCN block to refine the predictions from the previous stage further.

In previous studies, the DNN model [28] achieved a static RMSE of $0.259^\circ/\text{s}$ and dynamic RMSE of $0.143^\circ/\text{s}$ (measured in angular velocity), and the LSTM model [30] achieved an RMSE of 0.07° (using angular displacement and the details of the dataset is not reported). These results are provided for 1 step ahead prediction only. In our 5-step ahead prediction, we achieved a lower RMSE of 0.0137° (joint angular displacement). The CNN model [31] achieved an RMSE of $0.001^\circ/\text{s}$ using 10-step ahead prediction (measured in angular velocity).

Since the units and physical meaning of the error of previous studies differ from those of our work, it is only feasible to compare the error relative to the minimum and maximum values of prediction. Among these studies, only Ibrahim et al. [31] reported that the tremor predicted ranges from around $-25^\circ/\text{s}$ to $30^\circ/\text{s}$ (shown in a graph, without exact value). Our proposed method of having an RMSE of 0.0137° to measure within -10.70° to 14.41° is comparable to their performance. Nevertheless, interpretation of the performance with different studies necessitates the reported test conditions and study activities.

Owing to the above reasons, we have reproduced the baseline DL methods (i.e., DNN, LSTM, and CNN) and evaluated the performance with our proposed method using the same datasets. The results shown in Table 2 confirm that the proposed SADTCN model demonstrated superior performance than the baseline DL models.

While the performance of the proposed model is promising, it has some limitations for potential future work. Specifically, the self-attention mechanism provides interpretable attention weights; however, understanding the exact meaning and clinical significance of the attention weights might be challenging. The interpretation may vary depending on tasks and specific tremor characteristics. More advanced techniques, such as attention visualization and saliency mapping, may be required for comprehensive interpretability. On the other hand, while the model achieves low RMSE, it may not be directly translated into clinical decision-making without domain expert validation. Fine-tuning and adapting the model to specific clinical settings may require collaboration with medical professionals. Particularly, its effectiveness in real-world clinical practice requires rigorous validation and clinical trials. Implementing the model in a clinical setting may need to address the issues related to data privacy, regulatory compliance, and integration with existing medical systems.

5. Conclusion

In this paper, we have developed a SADTCN model for predicting hand-arm PDT signals in real-time from different activities. An experimental setup is established to record tremors in PD patients while engaged in various activities. The results demonstrated that the proposed model can accurately track a range of PDT frequencies and amplitudes from different activities and joint angular motions, and outperformed existing DL models for multi-step ahead predictions with different horizons. We analyzed the spectrum of predicted and

measured signals and found that the proposed model can accurately predict the transience of tremor amplitude in frequency and time. The Wilcoxon signed-rank test results show a statistically significant improvement of the proposed model over other DL models in all conditions.

Future work will focus on the integration of the developed model with wearable assistive devices to explore its ability to provide active tremor compensation in real-time. Advanced techniques, such as attention visualization and saliency mapping, will be applied to improve the interpretability of self-attention weights so that their clinical significance can be better understood. Moreover, domain-specific fine-tuning and validation in collaboration with medical professionals will aim to enhance the model's adaptability and relevance in clinical settings. Applying the proposed algorithm to other fields of research, such as the prediction of physiological tremors in surgeons to aid in microsurgery, would be an interesting future study.

CRedit authorship contribution statement

Guan Yuan Tan: Writing – original draft, Validation, Software, Methodology. **A.S.M. Bakibillah:** Writing – review & editing, Supervision, Project administration, Investigation, Conceptualization. **Ping Yi Chan:** Writing – review & editing, Supervision, Project administration. **Chee Pin Tan:** Writing – review & editing, Supervision, Resources, Project administration. **Surya Nurzaman:** Writing – review & editing, Supervision.

Ethics statement

The clinical tremor measurement was approved in July 2015 (protocol no. NMRR-14-1694-21740 [IIR]) by Medical Research Ethics Committee, Secretariat of the National Institutes of Health, Malaysia, and strictly adhered to the Guidelines from the National Institution of Health guidelines. All participants provided written informed consent to participate in this study.

Declaration of competing interest

None declared.

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Data availability

The dataset used and/or analyzed during the current study is available from the corresponding author upon reasonable request.

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